

ADI SINGHAL

+1 (929) 400-3809 • adis@nyu.edu • [linkedin.com/in/adi-singhal](https://www.linkedin.com/in/adi-singhal) • github.com/adityaarunsinghal

Applied AI Scientist & Engineer -- 4+ years shipping production RAG,
Agentic AI, and LLM-evaluation systems at Amazon (AWS Bedrock; Creators & Social)

WORK EXPERIENCE

Applied Scientist, Amazon Creators and Social, New York

Nov 2025 - Present

Selection & Quality Science · Agent Evaluation

- Built the Fraud Detection Multi-Agent System (VLM-powered browser agents investigating trademark violations on the open web), reaching 90% precision / 86% recall / 9% FPR (with the high-stakes "Ban Creator" action correct 94% of the time) -- productionized, and set the team's direction for the upcoming quarters.
- Prototyped a vision-language seasonality classifier (Claude on Bedrock; forced tool-use + prompt caching) with a two-stage harvest-then-score design that auto-built a 41-season taxonomy from 9k product images and scored a 15M+ ASIN catalog at 2¢/ASIN -- distinguishing marketplace-specific windows (e.g. hemisphere-inverted ski season) to gate out-of-season ads.
- Architected a nine-option science design space for selecting the best of N product-image variants (classical CV, fine-tuned ViT/ConvNeXt classifiers, VLMs, CLIP-embedding ranking, knowledge distillation), each costed; the chosen hybrid cascade cut inference cost 90%+ (\$1,120 vs \$10,500 per 100K ASINs) while keeping VLM-level accuracy on the hard cases.
- Filed a patent on the Selection Quality Scorer (SQS) pipeline -- a closed-loop, self-improving multimodal scorer that learns ad-worthiness from human judgment and scores a 55M-item catalog (in production as SQS v1/v2); invented its effort-weighted Dawid-Skene consolidation step that reweights 30 raters by category-specific purchase behavior (Spearman $\rho = 0.55$), re-adjudicating 5% of borderline labels by domain expertise.
- Improved the Ad Content Critic LLM-as-judge image-quality classifier by +25.2% F1 (to 0.79) on a held-out set with non-overlapping bootstrap CIs (stat sig) -- halving false alarms while catching more bad images at 0.4¢/image, via defect-reframing and an active-learning prompt optimizer (open-sourced).
- Ran a large-scale investigation of product-priorities (51-signal $\times$ 5-target sweep on 54M candidate ASINs); disproved the leadership "quality predicts ad revenue" hypothesis and shipped a human-grounded dataset of ad-worthiness to serve as the new organizational benchmark.

Software Engineer, Amazon Web Services, New York

July 2022 - Oct 2025

Bedrock RAG Platform · Agentic AI & Internal Tooling

- Built and evangelized BedrockBot, an internal Slack AI assistant now used by 25k+ Amazonians as the primary way to connect team context to Slack; designed its novel cross-account auth + Knowledge Base RAG integration. Won Amazon's "Deliver Results" award.
- Built OnCall Companion, an on-call automation agent on the Strands SDK + MCP (200+ Sev-2 tickets/week); adopted by 650+ engineers across 6+ teams, cutting MTTR 15-45 min/incident (30-80% faster, user-reported) with 26% fully automated RCA; grew a 25+ engineer maintainer community.
- Led Day-0 foundation-model support for Bedrock Knowledge Bases (managed RAG), cutting new-model onboarding from 5 days to 4 hours and scaling to 30-40+ models across 8 providers via zero-downtime, config-driven releases.
- Built an automated multi-model RAG benchmarking harness (16 datasets, 80 configs) scoring faithfulness, citation accuracy, and latency, enabling data-driven model selection at launch and end-to-end regression testing.
- Architected support for long-running reasoning-models, extending limits from 5-min/8k-token to 60-min/128k-token via hybrid DynamoDB/S3 session storage and a six-layer cascading-timeout design.
- Built a Tiktoken-based chunker that cut indexing/ingestion failures 20-25 $\times$ (95% to near-zero), eliminating the top customer ingestion complaint after 4 rounds of offline experiments on 10k real documents.
- Created the health.amazon.com subdomain, integrated One Medical sign-ups with Amazon, and founded the Health Services Applied Science Club. Pioneered Amazon Health's first conversational AI system (Project Macaw) in 3 weeks, achieving a 57% accuracy improvement over baseline in diagnosis prediction and presenting directly to SVP leadership (Neil Lindsay).

NLP Engineer, Kasisto Inc, New York

May 2021 - 2022

- Boosted the company's intent-detection model by 7% F1 (near-human) across 3 client deployments -- JPMC, TD Bank, and WestPac; tapped by leads to report directly to the CTO on model accuracy.
- Wrote Python to push 125+ experiments to the company HPC, presented the new pipeline and findings company-wide, and maintained rigorous experiment documentation for the Data Science team.
- Built a scalable framework for automatic model improvement by folding in gold-standard data (patent pending).

PROJECTS AND RESEARCH

NYU Agentic AI + MCP Workshop for Alumni

Fall 2025

- Ran a 4-week, classroom-style Model Context Protocol (MCP) workshop (interactive Python notebooks, final show-and-tell) for NYU graduate students and alumni.
- Built a production-ready MCP server for intelligent news aggregation using vector similarity search and multi-source integration, with an extensible template for new sources, handlers, and AI content analysis.
- Collaborated with a core contributor of MCP's Python SDK and released the learning module publicly on HackerNews (YC). Site: adityasinghal.com/agentic-ai-workshop · github.com/adityaarunsinghal/agentic-ai-workshop-2025

LLM-as-Judge Prompt Improver

May 2026

- Built an OPRO-style self-improving prompt-optimization framework (27k LOC Python + TypeScript) on Bedrock/Claude via Strands, with focus-pinned stratified k-fold CV, bootstrap CIs, SHA256-verified holdout discipline, and automatic rollback on F1 regression; 61 pytest files (14k test lines). github.com/adityaarunsinghal/LLM-As-A-Judge-Prompt-Improver

Meta AI / FAIR -- Computational Simulation of Perceived Difficulty in Games

May 2022

- Under FAIR researchers (Drs. Lake, Gureckis, Davidson), built a 2D-physics DSL + OpenAI-Gym framework over 64k+ automated scenarios; a 12-participant study found a strong negative correlation (-0.7) between human difficulty ratings and computational playability. github.com/adityaarunsinghal/temporal-goals-in-games

IBM -- Domain Adaptation

Summer 2021

- Built a hierarchical RoBERTa fine-tuning approach using domain-specific NER as an auxiliary task for zero-shot QA, beating baseline in 3 of 4 domains without target-domain QA data. github.com/adityaarunsinghal/Domain-Adaptation
- Published in *NAACL Workshop 2022*

Facebook Parl.ai -- Empathetic Chatbots

Fall 2020 - 2022

- Built "Empathetic Conversations in the Wild" (30k Reddit conversations, RoBERTa classifier), fine-tuned BlenderBot (9.4B) on HPC, and validated automated scoring vs human annotation ($r = 0.62$, $p < 4e-9$).

EDUCATION

B.A. Data Science, New York University -- GPA 3.98

Aug 2018 - May 2022

Advanced Statistics for Research • Computer Systems Org (Assembly) • Reinforcement Learning • Natural Language Processing • Responsible Data Science (Ethical AI) • High Performance Computing (HPC) • Causal Inference

B.A. Honors Psychology, New York University -- GPA 4.0

Aug 2018 - May 2022

Lab in Industrial/Organizational Psychology • Fundamentals of Decision-Making • Human Motivation & Volition • Computational Cognitive Neuroscience

ACHIEVEMENTS

- Org-wide "**AI Force Multiplier**" award for making cross-engineering primitives 2026
- "**Best Poster Award**" at the Amazon GenAI Symposium for Brand <> Creator Matching 2025
- Amazon's "**Deliver Results**" award for making a company-wide Slack AI Agent 2024
- Inducted to **Phi Beta Kappa** as a Junior 2021
- **Dean's Research Fund** recipient for Affective Computing research 2020

TECHNICAL SKILLS

GenAI / RAG: end-to-end production RAG, Amazon Bedrock (Retrieve/RetrieveAndGenerate/Converse/Agents, Knowledge Bases), multi-model orchestration (Claude, Llama, Mistral, Nova, Titan, DeepSeek), chunking/embeddings, hybrid & graph RAG (Neptune), vector DBs (OpenSearch/AOSS, FAISS).

Agentic AI: Bedrock AgentCore (browser tool, Runtime, Gateway), Strands SDK, Model Context Protocol (MCP), multi-agent/supervisor orchestration, tool use/function calling, prompt caching, agent eval & security (Guardrails, prompt-injection mitigation).

Evaluation / Science: LLM/VLM-as-judge design & calibration, held-out test design, bootstrap CIs, significance testing (Cohen's d, Mann-Whitney U), inter-rater reliability (Krippendorff's alpha, Cohen's kappa), active learning, causal & A/B design.

Platform / Languages: distributed systems, cascading-timeout design, event-driven pipelines (Streams→Lambda→SQS), cross-account IAM/STS, Step Functions, CDK/CloudFormation; Python, Java, TypeScript, SQL; PyTorch, HuggingFace, scikit-learn, pandas, LangChain, Pydantic; Docker, Kubernetes.